

1 Introductory Graph Theory

1.1 Degrees and Degree Sequences

Theorem 1.1. *Let G be a graph with size m . Then $\sum_{v \in V(G)} \deg v = 2m$.*

Corollary 1.2. *Every graph has an even number of vertices of odd degree.*

Theorem 1.3. *Let $d_1, d_2, \dots, d_n$ be a nonincreasing sequence of nonnegative integers. Then $(d_1, \dots, d_n)$ is graphic (degree sequence of some graph) if and only if $d_1 + \dots + d_n$ is even and*

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{d_i, k\}$$

for every $k = 1, \dots, n$.

Theorem 1.4. *Let $d_1, \dots, d_n$ be a nonincreasing sequence of nonnegative integers. Define $d_{n+1} = 0$. Then $(d_1, \dots, d_n)$ is graphic if and only if*

$$\sum_{i=1}^n d_i \text{ is even, and } \sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{d_i, k\}$$

for each $k = 1, \dots, n$.

1.2 Walks and Components

Theorem 1.5. *Every u - v walk in a graph contains a u - v path.*

1.3 Cut-sets and Blocks

Theorem 1.6. *Let G be a graph and let $v \in V(G)$. Then v is a cut-vertex of G if and only if there exist distinct vertices u and w of G such that every u - w path in G contains v .*

Theorem 1.7. *Let G be a graph and let $e \in E(G)$. Then e is a bridge of G if and only if there exist distinct vertices u and w of G such that every u - w path in G contains the edge e .*

Theorem 1.8. *Let G be a graph and let $e \in E(G)$. Then e is a bridge of G if and only if e is not contained in any cycle of G .*

Theorem 1.9. *A graph G of order at least 3 is a block (2-connected component) if and only if for every pair of vertices $u, v \in V(G)$ there is a cycle of G containing both u and v .*

Corollary 1.10. *A graph G of order at least 3 is a block if and only if for every pair of distinct vertices $u, v \in V(G)$ there exist at least two internally disjoint u - v paths in G .*

1.4 Distance in Graphs

Theorem 1.11. *Let G be a connected graph. Then*

$$\text{rad}(G) \leq \text{diam}(G) \leq 2\text{rad}(G).$$

1.5 Bipartite Graphs

Theorem 1.12. *A graph is bipartite if and only if it contains no odd cycle.*

1.6 Trees

Theorem 1.13. *Every non-trivial tree has at least two end-vertices (vertices of degree 1).*

Theorem 1.14. *The center of a tree is isomorphic to K_1 or K_2 .*

Theorem 1.15. *Let G be a connected graph of order n . Then G is a tree if and only if G has size $n - 1$.*

Corollary 1.16. *Every forest F of order n and with k components has size $n - k$.*

Theorem 1.17. *Let G be a graph of order n and size m . If any two of the following three statements hold, then G is a tree:*

1. G is connected,
2. G is acyclic,
3. $m = n - 1$.

Theorem 1.18. *Let T be a tree of order k , and let G be a graph with minimum degree $\delta(G) \geq k - 1$. Then T is a subgraph of G .*

Theorem 1.19. *Every connected graph G contains a spanning subgraph which is a tree (i.e. every connected graph has a spanning tree).*

Theorem 1.20. *Let G be a connected graph and run Kruskal's algorithm to obtain a spanning subgraph T . Then T is a minimum spanning tree of G (a spanning tree of minimal total weight).*

2 Connectivity

2.1 Vertex-cuts and Edge-cuts

Theorem 2.1. *Let G be an incomplete connected graph. If S is an edge-cut of G , then there is a vertex-cut S' of G such that $|S'| \leq |S|$.*

Theorem 2.2. (Whitney's Theorem) *Let G be a connected graph. Then*

$$\kappa(G) \leq \lambda(G) \leq \delta(G),$$

where $\kappa(G)$ is the vertex-connectivity, $\lambda(G)$ is the edge-connectivity, and $\delta(G)$ is the minimum degree.

2.2 k -Connectivity

Theorem 2.3. Let G be a graph of order $n \geq 2$, and let $k \in \{1, \dots, n-1\}$. If

$$\delta(G) \geq \left\lceil \frac{n+k-2}{2} \right\rceil,$$

then G is k -connected.

2.3 Menger's Theorem

Corollary 2.4 (Fan Lemma). Let G be a k -connected graph. If $u, v_1, \dots, v_k$ are any $k+1$ vertices of G , then there exist k internally disjoint paths $P_1, \dots, P_k$ such that P_i is a u - v_i path for each i .

Theorem 2.5 (Menger's Theorem). Let u and v be non-adjacent vertices in a connected graph G . Then the minimum number of vertices whose removal separates u and v equals the maximum number of pairwise internally disjoint u - v paths in G .

Theorem 2.6. (Whitney's Theorem on k -connectivity) A non-trivial graph G is k -connected if and only if for every pair of distinct vertices $u, v \in V(G)$ there are at least k internally disjoint u - v paths in G .

Theorem 2.7 (Dirac's Theorem). Let G be a k -connected graph and let S be any set of k vertices of G . Then G has a cycle that contains every vertex of S .

Theorem 2.8 (Győri–Lovász Theorem). Let G be a k -connected graph of order n , where $k \leq n-1$. If $v_1, \dots, v_k$ are distinct vertices of G and $n_1, \dots, n_k$ are positive integers with $n_1 + \dots + n_k = n$, then there exist k pairwise disjoint connected subgraphs $G_1, \dots, G_k$ of G such that $v_i \in V(G_i)$ and $|V(G_i)| = n_i$ for each $i = 1, \dots, k$.

3 Eulerian and Hamiltonian Graphs

3.1 Eulerian Graphs

Theorem 3.1. A non-trivial connected graph G has an Eulerian circuit (a closed trail containing all edges exactly once) if and only if every vertex of G has even degree.

Theorem 3.2. A non-trivial connected graph G has an Eulerian trail (an open trail containing all edges exactly once) if and only if G contains exactly two vertices of odd degree (the trail must begin and end at those two vertices).

3.2 Hamiltonian Graphs

Theorem 3.3. Let G be a Hamiltonian graph of order n . Then for every proper nonempty subset $S \subset V(G)$, the number of connected components of $G - S$ satisfies $k(G - S) \leq |S|$.

Theorem 3.4 (Ore's Theorem). *Let G be a graph of order $n \geq 3$. If for every pair of distinct non-adjacent vertices u, v we have $\deg(u) + \deg(v) \geq n$, then G is Hamiltonian.*

Corollary 3.5. *Let G be a graph of order n and u, v distinct non-adjacent vertices with $\deg(u) + \deg(v) \geq n$. Then G is Hamiltonian if and only if $G + uv$ (adding the edge uv) is Hamiltonian.*

Theorem 3.6 (Chvátal–Erdős Theorem). *Let G be a graph of order $n \geq 3$. If the vertex-connectivity $\kappa(G)$ is at least the independence number $\alpha(G)$, then G is Hamiltonian.*

4 Planar Graphs

4.1 Jordan's Theorem and Euler's Formula

Theorem 4.1 (Jordan Curve Theorem). *A simple closed curve in the plane divides the plane into two regions, exactly one of which is bounded, with the curve as the common boundary of both.*

Theorem 4.2 (Euler's Formula). *Let G be a connected plane graph with n vertices, m edges, and f faces. Then*

$$n - m + f = 2.$$

4.2 Maximal Planar Graphs

Lemma 4.3. *Let G be a maximal plane graph of order at least 3, and let F be any face of G . Then:*

1. *The boundary of F in G is a triangle (a 3-cycle).*
2. *If two edges xy and xz lie on the boundary of F , then yz is also an edge of G .*

Theorem 4.4. *If G is a maximal planar graph of order $n \geq 3$ and size m , then*

$$m = 3n - 6.$$

Lemma 4.5. *If G is any planar graph, then $\delta(G) \leq 5$ (there is a vertex of degree at most 5).*

Theorem 4.6. *Let u be a vertex in a maximal planar graph G of order at least 4. Then there is a unique cycle in G whose vertex set is $N(u)$ (the neighbors of u).*

Theorem 4.7. *Let G be a maximal planar graph of order at least 4. Then*

$$3 \leq \delta(G) \leq 5.$$

Theorem 4.8. *If S is a minimal separating set in a maximal planar graph G of order at least 4, then the subgraph $G[S]$ has minimum degree at least 2.*

Corollary 4.9. *If G is a maximal planar graph of order at least 4, then $3 \leq \kappa(G) \leq 5$ (vertex-connectivity between 3 and 5).*

4.3 Chvátal's Art Gallery Theorem and Fáry's Theorem

Theorem 4.10 (Chvátal's Art Gallery Theorem). *If P is a simple polygon with n vertices, then there is a set S of at most $\lfloor n/3 \rfloor$ vertices of P such that every point of P is visible from at least one vertex in S .*

Corollary 4.11. *If P is a simple polygon with n vertices, then there is a set S of at most $\lfloor n/3 \rfloor$ points in the interior of P such that every point of P is visible from some point in S .*

Theorem 4.12 (Fáry's Theorem). *Every planar graph has a straight-line embedding in the plane (can be drawn with edges as straight segments).*

Theorem 4.13 (Four Color Theorem). *Every planar graph is 4-colorable (its vertices can be properly colored with at most 4 colors).*

Theorem 4.14 (Five Color Theorem). *Every planar graph is 5-colorable.*

5 Graph Colourings

5.1 Introduction to vertex colourings

Theorem 5.1. *Let G be a graph of size m . Then the chromatic number $\chi(G) \leq \frac{1}{2}(1 + \sqrt{1 + 8m})$.*

Theorem 5.2. *Let G be a graph. Then $\chi(G) \leq \Delta(G) + 1$.*

5.2 Critically and minimally k -chromatic graphs

Theorem 5.3. *Let G be a graph. The following statements hold:*

1. *If G contains a cut-vertex, then G cannot be critically k -chromatic.*
2. *If G contains a cut-vertex, then G cannot be minimally k -chromatic.*
3. *If G has no isolated vertices and is minimally k -chromatic, then G is critically k -chromatic.*

Theorem 5.4. *Let G be a critically k -chromatic graph. Then G is $(k-1)$ -edge-connected.*

Corollary 5.5. *Let G be a connected, minimally k -chromatic graph. Then G is $(k-1)$ -edge-connected.*

Corollary 5.6. *Let G be either critically k -chromatic or connected and minimally k -chromatic. Then $\delta(G) \geq k-1$.*

5.3 Brooks' Theorem

Theorem 5.7 (Brooks' Theorem). *Let G be a connected graph which is neither a complete graph nor an odd cycle. Then $\chi(G) \leq \Delta(G)$.*

Theorem 5.8. *For every graph G ,*

$$\chi(G) \leq 1 + \max\{\delta(H) : H \text{ is an induced subgraph of } G\}.$$

5.4 Further results on vertex-colourings

Theorem 5.9. *For every positive integer k , there exists a triangle-free graph with chromatic number k .*

Theorem 5.10 (Nordhaus–Gaddum Theorem). *Let G be a graph of order n . Then*

$$2\sqrt{n} \leq \chi(G) + \chi(\overline{G}) \leq n + 1,$$

and

$$n \leq \chi(G)\chi(\overline{G}) \leq \left(\frac{n+1}{2}\right)^2,$$

where $\overline{G}$ is the complement of G .

5.5 Vizing's Theorem

Theorem 5.11 (Vizing's Theorem). *Let G be a graph. Then*

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1,$$

where $\chi'(G)$ is the edge-chromatic number of G .

5.6 Further results on edge-colourings

Theorem 5.12. *Let G be a nonempty graph in which all vertices of maximum degree are independent. Then G is class one (its edges can be colored with $\Delta(G)$ colors).*

Theorem 5.13. *Let G be a graph of size m . If $m > \Delta(G) \cdot \beta'(G)$, then G is class two (requires $\Delta(G) + 1$ edge-colors).*

6 Covers and Matchings

6.1 Covers

Theorem 6.1. *Let G be a graph. Then $\alpha(G) + \beta(G) = |V(G)|$, where $\alpha(G)$ is the vertex-independence number and $\beta(G)$ is the vertex-cover number.*

Theorem 6.2. *Let G be a graph without isolated vertices. Then $\alpha'(G) + \beta'(G) = |V(G)|$, where $\alpha'(G)$ is the matching number and $\beta'(G)$ is the edge-cover number.*

Theorem 6.3. *Let G be any graph. Then $\alpha(G) \geq \beta'(G)$.*

6.2 Matchings

Theorem 6.4. *Let G be a bipartite graph. Then $\alpha(G) = \beta'(G)$ (the size of a maximum matching equals the size of a minimum vertex cover).*

Theorem 6.5 (Hall's Marriage Theorem). *Let G be a bipartite graph with bipartition (A, B) . Then G contains a matching that matches every vertex of A if and only if for every subset $S \subseteq A$ we have $|N(S)| \geq |S|$.*

Theorem 6.6. *Let G be a $2k$ -regular graph of even order. Then G has a 2-factor (a spanning 2-regular subgraph).*

Corollary 6.7. *Let G be a bipartite graph with bipartition (A, B) . If $|N(S)| \geq |S| - d$ for every subset $S \subseteq A$ (for some integer $d \geq 0$), then G contains a matching of size $|A| - d$.*

Corollary 6.8. *Let G be a k -regular bipartite graph with $k \geq 1$. Then G has a perfect matching.*

6.3 Tutte's Theorem

Theorem 6.9 (Tutte's Theorem). *A graph G has a 1-factor (perfect matching) if and only if for every subset $S \subseteq V(G)$, the number of odd components of $G - S$ is at most $|S|$.*

Theorem 6.10. *Every bridgeless cubic graph has a 1-factor.*

7 Group Actions

7.1 Orbits and Stabilizers

Lemma 7.1. *Let Γ be a finite group acting on a set S , and let $x, y \in S$ be similar (i.e. $y = x\alpha$ for some $\alpha \in \Gamma$). Then:*

1. *The set of elements of Γ that map x to y is a right coset of Γ_x (the stabilizer of x).*
2. *If α, β lie in the same right coset of Γ_x , then $x\alpha = x\beta$.*
3. *The stabilizers of x and y are conjugate subgroups in Γ . Moreover, if $y = x\alpha$, then $\Gamma_y = \alpha^{-1}\Gamma_x\alpha$.*

Theorem 7.2 (Orbit-Stabilizer Theorem). *Let Γ be a finite group acting on a set S . Then for all $x \in S$,*

$$|\Gamma_x| \cdot |x\Gamma| = |\Gamma|,$$

where Γ_x is the stabilizer of x and $x\Gamma$ is the orbit of x .

7.2 Burnside's Lemma

Theorem 7.3 (Cauchy–Frobenius (Burnside) Lemma). *Let Γ be a finite group acting on a finite set S . Then the number of orbits of S under the action of Γ is*

$$\frac{1}{|\Gamma|} \sum_{\alpha \in \Gamma} |\text{Fix}(\alpha)|,$$

where $\text{Fix}(\alpha) = \{x \in S : x\alpha = x\}$ is the set of points fixed by α .